

THE THREE RULES 1. Peripheral IV cannula is the default route for standard lymphoma regimens. 2. PICC where the scored proforma indicates it. 3. PICC is mandatory for continuous vesicant infusion of 24 hours or more.

1. Scored route selection
Complete for every patient before starting IV chemotherapy. Score each domain and add A + B + C + D. The total score determines the recommendation. Italic letters indicate the evidence source — see the legend on page 2.

DOMAIN A — Regimen Vesicancy		Score
☐	Non-vesicant regimen — Rituximab, Bendamustine, Obinutuzumab, Chlorambucil	0
☐	Irritant agents only — Cyclophosphamide alone, Dacarbazine alone [A, F]	1
☐	Non-DNA vesicant — Vincristine, Bleomycin, Brentuximab vedotin, Polatuzumab vedotin (in R-CHOP, BV-AVD, Pola-R-CHP, R-CVP, O-CVP) [A, F]	2
☐	DNA-binding vesicant — Doxorubicin or Daunorubicin (in ABVD, R-CHOP, Pola-R-CHP, BV-AVD, EPOCH, AVD) [A, F]	3
★	STOP — Continuous vesicant infusion ≥24 hours (DA-R-EPOCH 96 h pump). PICC is MANDATORY . Do not proceed with PIV. [E]	PICC

DOMAIN B — Venous Access Quality		Score
☐	Good — reliable, easily visible veins, cannulated without difficulty	0
☐	Fair — achievable but requires skill; thin or partially visible veins	1
☐	Fragile — barely visible, high-risk for extravasation injury on cannulation [A, B]	2
☐	DIVA — two or more failed peripheral cannulation attempts on this occasion, formally documented [B, C]	3

DOMAIN C — Patient Risk Factors		Add
Tick all that apply, +1 each, maximum 4.		
☐	Prior extravasation on this or an equivalent vesicant regimen — documented in the clinical record [A, B]	+1
☐	Cannot reliably report extravasation symptoms — documented cognitive impairment, severe dementia, or communication disorder. Age alone does NOT qualify. [A, E]	+1
☐	Documented significant skin or vein fragility beyond normal ageing — e.g. severe steroid-induced skin thinning, clinically assessed [A, B]	+1
☐	Informed patient preference for PICC — after explicit disclosure of both the PIV extravasation risk and the 7.1% PICC-related clot risk in lymphoma patients. Both disclosed and documented. Where a locally approved PICC patient-information leaflet covering extravasation, infection and thrombosis forms part of signed pre-procedure consent, this constitutes the documented patient education requirement for PICC insertion. [D]	+1

DOMAIN D — Monitoring Capacity		Score
Can continuous observation of the PIV insertion site be ensured throughout the entire vesicant bolus for this patient on this occasion? [A, B]		
☐	YES — continuous observation can be guaranteed for this patient on this occasion	0
★	NO — continuous observation cannot be ensured. This overrides the total score and requires PICC regardless of domains A + B + C.	PICC

TOTAL SCORE (A + B + C + D) → DECISION			
Score	Decision	Action required	Document
0 - 2	PIV Appropriate	Proceed with PIV. Apply all 8 mandatory precautions (Section 3). Standard documentation.	Score in notes
3 - 5	PIV with Enhanced Precautions	Proceed with PIV. Senior clinician or nurse present throughout. All precautions mandatory. Countersignature required at scores 4–5.	Score + reason + countersign
≥6 or D = No	PICC Indicated	Do not administer vesicant via PIV. Complete the pre-PICC checklist (Section 2). Refer to the vascular access team. Escalate to consultant if contraindicated.	Score + criterion + escalation

Patient: _____ ID: _____ Date: _____

A = ____ B = ____ C = ____ D = ____ **Total = ____** Route: PIV ☐ PICC ☐

2. Pre-PICC insertion safety checklist

Complete before referring for PICC insertion. Items 1–5 are absolute requirements. Items 6–7 are relative — apply clinical judgement.

✓	Requirement	Evidence	Result / notes
☐	Platelets ≥10×10 ⁹ /L	[C, E]	Result: _____ Date: _____
☐	No active infection at proposed insertion site — inspect both arms	[C]	Pass ☐ Fail ☐
☐	No bilateral upper limb deep vein thrombosis	[C]	Pass ☐ Fail ☐
☐	7.1% PICC-related clot risk in lymphoma disclosed to patient — document explicitly, this is lymphoma-specific	[D]	Signed: _____
☐	No severe uncorrectable coagulopathy — INR <3.0 or corrected	[C]	Pass ☐ Fail ☐
☐	No ipsilateral axillary clearance (relative — use contralateral arm if present)	[C]	
☐	Renal function considered — CKD stage 3B or worse (relative — consider preserving veins for future dialysis access)	[C]	

Referring clinician: _____ Signature: _____ Date: _____

Consultant countersignature (proforma score 4–5): _____ Date: _____

3. Mandatory PIV precautions — all vesicant administration

All 8 steps are mandatory whenever a vesicant drug is given by peripheral IV. The published 0.05% extravasation rate per infusion day [G] reflects what structured PIV administration looks like in practice.

#	Action	Detail	✓
1	Patency check — before starting [A, B]	Flush 10 mL saline and confirm blood return before any drug. If absent, or if there is resistance — do not proceed.	☐
2	Site selection [A, B]	Forearm vein preferred (cephalic or basilic). Avoid hand, wrist, antecubital fossa, joint areas, prior bruising or phlebitis.	☐
3	Patient education — before infusion [A, B, I]	Tell the patient: “Tell us immediately if you feel burning, stinging, swelling, or any change at the drip site.” Also inform the patient about hypersensitivity reactions: flushing, rash, breathlessness, chest tightness, or any sudden change in how they feel during the infusion. Both discussions must be documented as completed at the first visit.	☐
4	Continuous observation during bolus [A, B]	Observe the infusion site throughout the entire bolus. Do not leave the patient. If this cannot be ensured → Domain D = No → PICC required.	☐
5	Extravasation kit — confirm before starting [A, B, F]	Confirm the kit is present and accessible before starting. Dexrazoxane for doxorubicin must be administered within 6 hours of extravasation.	☐
6	Flush between agents [A]	10 mL saline between each vesicant agent. Never co-administer two vesicants without an intervening flush.	☐
7	Documentation [B, I, J, K]	Record in the chemotherapy notes: cannula site, gauge, patency check result, VIP score (0–5) at each cannula assessment, patient education and hypersensitivity discussion given, and monitoring performed. VIP score ≥2 requires cannula removal before continuing.	☐
8	Reassess at every cycle [B, H]	If access quality deteriorates, or any prior cycle complication occurred, repeat the proforma before the next treatment.	☐

Evidence base

[A] Pérez Fidalgo JA et al. Management of chemotherapy extravasation: ESMO-EONS clinical practice guidelines. Ann Oncol 2012;23(Suppl 7):vii167–vii173. doi:10.1093/annonc/mds294

[B] Barton A. NIVAS Infiltration and Extravasation Toolkit. National Infusion and Vascular Access Society. February 2024. nivas.org.uk

[C] Johnston AJ et al. Association of Anaesthetists guidelines: safe vascular access 2025. Anaesthesia 2025;80(11):1381–1396. doi:10.1111/anae.16727

[D] Zhang X, Huang JJ, Xia Y, et al. Oncotarget 2016;7(23):35404–35411. doi:10.18632/oncotarget.8639 — 7.1% PICC-related DVT in lymphoma (OR 3.849, p<0.001, n=8,028)

[E] Major A, Chopra V et al (MAGIC-ONC). Ann Intern Med 2025;178(12_Suppl):S143–S177. doi:10.7326/ANNALS-25-02523

[F] Matsumoto K, Ryushima Y, Sato J, et al (JSCN/JSMO/JASPO). ESMO Open 2024;9(10):103932. doi:10.1016/j.esmoop.2024.103932

[G] Tiger YK, Glennan P, Nedved A, Lukowski J, et al. Blood 2025;146(Suppl 1):518 (ASH 2025 abstract). doi:10.1182/blood-2025-518 — 0.05% PIV extravasation per infusion day with correct technique

[H] Audit 3354 (2025) — local service evaluation, n=32 evaluable.

[I] Jackson A. A battle in vein: infusion phlebitis. Nurs Times 1998;94(4):68–71.

[J] Gallant P, Schultz AA. Evaluation of a visual infusion phlebitis scale for determining appropriate discontinuation of peripheral intravenous catheters. J Infus Nurs 2006;29(6):338–345. doi:10.1097/00129804-200611000-00004

[K] Tzolos E, Salawu A. Improving the frequency of visual infusion phlebitis (VIP) scoring on an oncology ward. BMJ Qual Improv Rep 2014;3(1):u205455.w2364. doi:10.1136/bmjquality.u205455.w2364